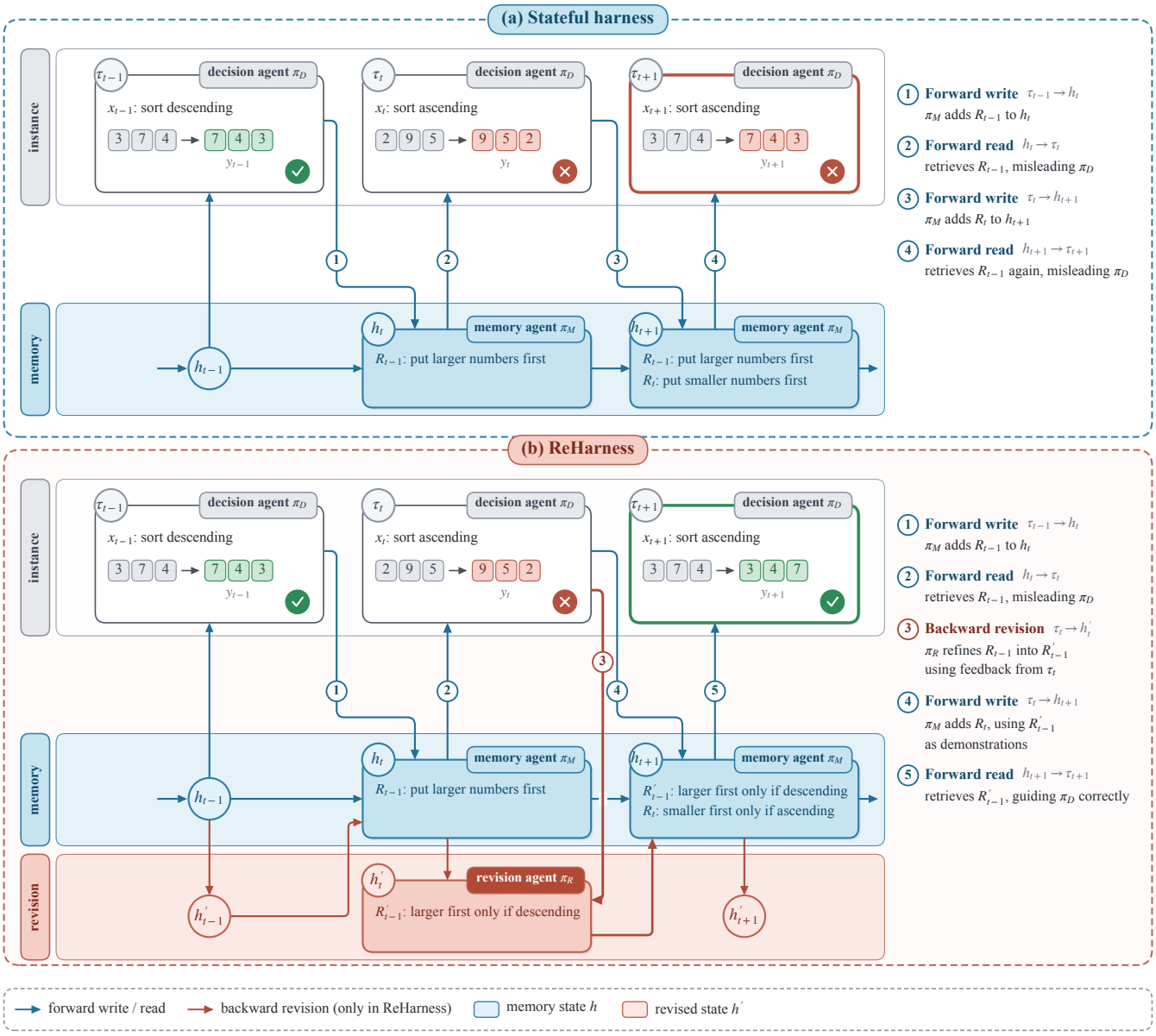

(a) Stateful harness
instance
memory
decision agent π_D
x_{t-1}: sort descending
3 7 4 → 7 4 3
y_{t-1}
x_t: sort ascending
2 9 5 → 9 5 2
y_t
x_{t+1}: sort ascending
3 7 4 → 7 4 3
y_{t+1}
memory agent π_M
R_{t-1}: put larger numbers first
R_t: put smaller numbers first
1 Forward write $\tau_{t-1} \to h_t$
π_M adds R_{t-1} to h_t
2 Forward read $h_t \to \tau_t$
retrieves R_{t-1}, misleading π_D
3 Forward write $\tau_t \to h_{t+1}$
π_M adds R_t to h_{t+1}
4 Forward read $h_{t+1} \to \tau_{t+1}$
retrieves R_{t-1} again, misleading π_D
(b) ReHarness
instance
memory
revision
decision agent π_D
x_{t-1}: sort descending
3 7 4 → 7 4 3
y_{t-1}
x_t: sort ascending
2 9 5 → 9 5 2
y_t
x_{t+1}: sort ascending
3 7 4 → 3 4 7
y_{t+1}
memory agent π_M
R_{t-1}: put larger numbers first
R'_{t-1}: larger first only if descending
R_t: smaller first only if ascending
revision agent π_R
R'_{t-1}: larger first only if descending
1 Forward write $\tau_{t-1} \to h_t$
π_M adds R_{t-1} to h_t
2 Forward read $h_t \to \tau_t$
retrieves R_{t-1}, misleading π_D
3 Backward revision $\tau_t \to h'_t$
π_R refines R_{t-1} into R'_{t-1}
using feedback from τ_t
4 Forward write $\tau_t \to h_{t+1}$
π_M adds R_t, using R'_{t-1}
as demonstrations
5 Forward read $h_{t+1} \to \tau_{t+1}$
retrieves R'_{t-1}, guiding π_D correctly
forward write / read
backward revision (only in ReHarness)
memory state h
revised state h'